

STAT3003: Survey Methods

final review

Final exam

- 120 minutes
- close-book, but allow 1 page note (A-4, double side, any font size, either hand-writing or printing is fine)
- Similar format as the mid-term exam

Main Contents

- survey methods: SRS, stratified random sampling, cluster sampling (and two-stage cluster sampling), systematic sampling (and repeated systematic sampling)
- Ratio estimator for SRS, cluster sampling and two-stage cluster sampling
- 3 parameters: total τ , mean μ and proportion p
- 4 quantities: $E(\hat{\theta})$, $\widehat{Var}(\hat{\theta})$, $C.I.$, n

- Check midterm review

Systematic Sampling for τ : $\hat{\tau}$

- Estimator: $\hat{\tau}_{sys} = NY_1$, where $Y_i = \sum_{j=1}^{M_i} y_{ij}$ is total of observations in the i -th cluster
- $E(\hat{\tau}) = \tau$
- cannot calculate $\widehat{Var}(\hat{\tau}_{sys})$
- cannot calculate C.I.
- But if we can assume it is SRS

$$\Rightarrow \widehat{Var}(\hat{\tau}_{sys}) \approx \frac{M^2}{M_1} \left(1 - \frac{M_1}{M}\right) \frac{1}{M_1 - 1} \sum_{j=1}^{M_1} (Y_{1j} - \bar{Y}_1)^2$$

But if we can NOT assume it is SRS

- Therefore, we can see
 - $ICC \approx 0 \Rightarrow Var(\hat{t}_{SRS})$ estimates the true variance well
 - $ICC \gg 0 \Rightarrow Var(\hat{t}_{SRS})$ underestimates the true variance
 - $ICC \ll 0 \Rightarrow Var(\hat{t}_{SRS})$ overestimates the true variance
- In the setting of systematic sampling, which kinds of populations will have $ICC \approx 0, \gg 0$ or $\ll 0$?
 - It all depends how the sampling frame is ordered.

Repeated Systematic Sampling

- Repeated systematic sampling is essentially equivalent to cluster sampling
- Thus all the formulae from cluster sampling apply
 - $\hat{\tau} = N \frac{1}{n} \sum_{i=1}^n Y_i$
 - $Var(\hat{\tau}) = \frac{N-n}{N-1} \frac{1}{n} \sigma_c^2$, where σ_c^2 is the variance of cluster totals
 - $\widehat{Var}(\hat{\tau}) = N(N-n) \frac{1}{n} \hat{\sigma}_c^2$, where $\hat{\sigma}_c^2$ is the sample variance of the cluster totals
 - A $100(1 - \alpha)\%$ is given by $\hat{\tau} \pm t_{n-1, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\tau})}$

Ratio Estimators for SRS

- for ratio estimation of μ_y :
 - $\hat{\mu}_r = \frac{\bar{Y}}{\bar{X}} \mu_x, Var(\hat{\mu}_r) \approx \frac{N-n}{N-1} \frac{1}{n} \sigma_r^2, \widehat{Var}(\hat{\mu}_r) = \left(1 - \frac{n}{N}\right) \frac{1}{n} \hat{\sigma}_r^2$
- For estimation of τ_y we will use
 - $\hat{\tau}_r = \frac{\bar{Y}}{\bar{X}} \tau_x, Var(\hat{\tau}_r) \approx \frac{N-n}{N-1} \frac{N^2}{n} \sigma_r^2, \widehat{Var}(\hat{\tau}_r) = (N - n) \frac{N}{n} \hat{\sigma}_r^2$
- For estimation of R we will use
 - $\hat{R} = \frac{\bar{Y}}{\bar{X}}, Var(\hat{R}) \approx \frac{1}{\mu_x^2} \frac{N-n}{N-1} \frac{1}{n} \sigma_r^2, \widehat{Var}(\hat{R}) = \frac{1}{\mu_x^2} \left(1 - \frac{n}{N}\right) \frac{1}{n} \hat{\sigma}_r^2$
- where
 - $\sigma_r^2 = \frac{1}{N} \sum_{i=1}^N (y_i - Rx_i)^2$ and $\hat{\sigma}_r^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \hat{R}X_i)^2$

Ratio Estimators for SRS

- All biased
- Confidence intervals will be in the familiar form of
 - *Point Estimate $\pm t_{n-1, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(Estimator)}$*
- To find required sample sizes to achieve a $100(1 - \alpha)\%$ C.I. with width $2d$ we can either solve
 - $d = z_{1-\frac{\alpha}{2}} \sqrt{Var(Estimator)}$ for n **or**
 - $d = t_{n-1, 1-\frac{\alpha}{2}} \sqrt{Var(Estimator)}$ for n
 - Note both methods require some reasonable estimate of σ_r^2

When to use ratio estimator

- does not know the population size N
- or
- if $\hat{\rho} \gg \frac{1}{2} \frac{\left(\frac{\hat{\sigma}_x}{\bar{X}}\right)}{\frac{\hat{\sigma}_y}{\bar{Y}}}$, then $\hat{\sigma}_r^2 \ll \hat{\sigma}_y^2$

Ratio Estimation for Cluster Sample

- The (biased) ratio estimator for τ under cluster sampling is

- $$- \hat{\tau}_r = \frac{\bar{Y}}{\bar{M}} M = \hat{R} M, \text{ where } M = \sum_{j=1}^N M_j \text{ and } \bar{M} = \frac{1}{n} \sum_{i=1}^n M_i$$

- (We've changed notation here. Now $\bar{M}$ means the average cluster size in the sample, not in the population)

- $$- \widehat{Var}(\hat{\tau}_r) = N(N - n) \frac{1}{n} \hat{\sigma}_r^2 \text{ where } \hat{\sigma}_r^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \hat{R} M_i)^2$$

Ratio Estimation for Two-Stage Cluster Sample

- Recall $M_i \frac{1}{m_i} \sum_{j=1}^{m_i} y_{ij} = M_i \bar{Y}_i$ is an estimator for the total value in cluster i , i.e. $M_i \bar{Y}_i = \hat{Y}_i$
 - $\Rightarrow \frac{\sum_{i=1}^n M_i \bar{Y}_i}{\sum_{i=1}^n M_i} = \frac{\sum_{i=1}^n \hat{Y}_i}{\sum_{i=1}^n M_i} := \hat{\mu}_r$ is a reasonable estimator for μ
 - It is also a ratio estimator
- Because the sampling is two stage, $\widehat{Var}(\hat{\mu}_r)$ has two parts
 - $\widehat{Var}(\hat{\mu}_r) = \frac{1}{M^2} N (N - n) \frac{1}{n} \hat{\sigma}_r^2 + \frac{1}{M^2} \frac{N}{n} \sum_{i=1}^n M_i (M_i - m_i) \frac{1}{m_i} \hat{\sigma}_i^2$
 - Where $\hat{\sigma}_r^2 = \frac{1}{n-1} \sum_{i=1}^n (\hat{Y}_i - \hat{\mu}_r M_i)^2$ and $\hat{\sigma}_i^2$ is the sample variance of Y_{ij} inside cluster i

Regression Estimation

- $\hat{\mu}_L := a + b\mu_x$ where
 - $b = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$
 - $a = \bar{Y} - b\bar{X}$
- Much like the ratio estimator, **the regression estimator is biased under SRS**
 - But this bias is very small for large samples
- We may estimate the MSE of $\hat{\mu}_L$ by
 - $\widehat{Var}(\hat{\mu}_L) = \frac{N-n}{Nn} \frac{1}{n-2} \sum_{i=1}^n (Y_i - a - bx_i)^2$
 - where a and b are as before
- An approximate $(1 - \alpha)100\%$ C.I. for μ_y is
 - $\hat{\mu}_L \pm t_{n-2, 1-\frac{\alpha}{2}} \sqrt{\widehat{Var}(\hat{\mu}_L)}$
 - Notice the degrees of freedom

Survey errors

- Errors of Non-observation
 - Sampling error, error of coverage, non-response, reduce refusal rates,
- Errors of Observation
 - Errors due to the interviewer; respondent; measurement instrument; method of data collection
- Reducing Errors
 - Callbacks, rewards, incentives, interviewer training, data checks

Questionnaire

- Question Design
 - Question order, leading questions, order of responses, closed v open questions, “don’t know” response, middle-ground options, question balance; argument; tone; double-barrelledness; clarity, errors of memory
- Likert Scales
 - Likert items, Likert scales, statements, response scale, midpoint, British Social Attitudes survey, internal consistency, Cronbach’s α , balance

Interpreting Results

- **Confounding variables**
 - factors that have not been considered by researchers, but affect the suspected causes and the effect
- **Simpson's Paradox**
 - A trend that appears in groups of data can disappear or even reverse when the groups are combined

Random Response Model

for asking sensitive questions

- More generally, define
 - $p_S := \Pr(\text{sensitive qu.})$
 - $p_{T,Yes} := \Pr(\text{Yes}|\text{trivial qu.})$
- Then we have shown
 - $\phi = pp_S + p_T(1 - p_S)$
 - $p = \frac{\phi - p_T(1 - p_S)}{p_S} \Rightarrow \hat{p} := \frac{\hat{\phi} - p_T(1 - p_S)}{p_S}$
 - $\widehat{Var}(\hat{p}) = \frac{\widehat{Var}(\hat{\phi})}{p_S^2}$.
 - This shows the **penalty** for using the random response model: the variance is larger
- We could make p_S larger to reduce this problem, but the respondent may think the interviewer will know which question is being asked

Post-Stratification

- More generally, for a population with L strata, with N_i and $\bar{Y}_i$ are the population size and sample mean in strata i respectively, the **post-stratified estimator for μ** is

$$- \bar{Y}_{pst} = \frac{N_1}{N} \bar{Y}_1 + \frac{N_2}{N} \bar{Y}_2 + \cdots + \frac{N_L}{N} \bar{Y}_L$$

– The formulae for τ and p quickly follow

- $\widehat{Var}(\bar{Y}_{pst}) := \sum_{i=1}^L \frac{N-n}{nN} \frac{N_i}{N} \hat{\sigma}_i^2 + \sum_{i=1}^L \frac{1}{n^2} \frac{N-n}{N-1} \left(1 - \frac{N_i}{N}\right) \hat{\sigma}_i^2$

Subpopulations

- Thus a sensible estimator for μ_1 is

$$- \hat{\mu}_1 := \bar{Y}_1 = \frac{1}{n_1} \sum_{i=1}^{n_1} Y_{1i}$$

$$- \Rightarrow \widehat{Var}(\bar{Y}_1) = \frac{N^2}{N_1^2} \left(1 - \frac{n}{N}\right) \frac{1}{n} \frac{n_1-1}{n_1-1} \hat{\sigma}_1^2$$

– If $\frac{N_1}{N}$ is unknown, we plug in $\frac{n_1}{n}$ in its place

$$- \Rightarrow \widehat{Var}(\bar{Y}_1) \approx \left(1 - \frac{n}{N}\right) \left(\frac{n}{n_1-1}\right) \frac{n_1-1}{n_1} \frac{1}{n_1} \hat{\sigma}_1^2$$

– And if n and n_1 are large, $\widehat{Var}(\bar{Y}_1) \approx \left(1 - \frac{n}{N}\right) \frac{\hat{\sigma}_1^2}{n_1}$

Final Schedule

- Final exam on Dec. 10th , 6:30pm
- Homework 4: due on Dec. 4th at 9:30am, solution will be posted on Dec. 5th
- Project: due on Dec. 2nd at 9:30am
- Final QA:
 - ✓ welcome to email your questions to RA or me at any time (for fastest response);
 - ✓ you can also post your question on Discussion Board/online forum (for fast response);
 - ✓ I will also be in office on Dec. 10th from 9am-11:30am (for face-to-face QA)